
Calibrating and validating the loop against real fleet telemetry

The simulator has one feedback cycle: work → occupancy → power → temperature → throttle → clock → next interval's work. Each arrow is one equation with a few constants. This deck says which instrument on a real fleet measures each arrow, how its constants are fitted, and what must hold in real data before any fault label is trusted.

5

ARROWS IN THE
LOOP

14

CONSTANTS TO
FIT

3

INSTRUMENTS: DCGM, NSIGHT, THE
JOB

0

LABELLED INCIDENTS NEEDED FOR STAGES
1-2

Every arrow in the loop has a real instrument

Fleet-wide signals come from DCGM at about 1 Hz and are always on. Per-kernel truth comes from Nsight and is a one-off lab measurement per application. Timing comes from the job itself.

Arrow	Model term & constants	Real signal	Scope
work → occupancy	$occ = cells / (cells + c\%$ $c\% = 250 \text{ k cells}$	DCGM_FI_PROF_SM_ACTIVE, PIPE_FP64_ACTIVE, DRAM_ACTIVE; Nsight Compute achieved occupancy & SM throughput	fleet 1 Hz LAB per kernel
occupancy → power	$P = P_{idle} \cdot leak + (P_{max} - P_{idle}) \cdot occ \cdot f^{2.2}$ 70 W, 700 W, exponent 2.2	DCGM_FI_DEV_POWER_USAGE, ENFORCED_POWER_LIMIT; clocks locked with <code>nvidia-smi -lgc</code>	fleet + one clock sweep
power → temperature	$dT/dt = (P \cdot R_{th} - (T - T_{inlet})) / \tau$ $R_{th} 0.06 \text{ } ^\circ\text{C/W}$, $\tau 20 \text{ s}$, $0.9 \text{ } ^\circ\text{C/kW}$	DCGM_FI_DEV_GPU_TEMP; inlet from the rack BMC or cold-aisle sensor; rack power from the PDU	fleet + one step test
temperature → throttle	engage at 84 °C, HW at 90 °C, 3 °C hysteresis, 0.020 clock/°C	DCGM_FI_DEV_CLOCK_THROTTLE_REASONS bitmask; SLOWDOWN_TEMP, SHUTDOWN_TEMP read from the device	fleet read-only
throttle → clock	1755 MHz base, 1050 min, quantised to 15 MHz steps	DCGM_FI_DEV_SM_CLOCK; lattice from <code>nvidia-smi -q -d SUPPORTED_CLOCKS</code>	fleet read-only
clock → next work	compute $\propto 1 / f$, next interval; 88.7 ns per cell	the solver's per-rank phase timing (NVTX ranges under Nsight Systems, or its log)	job per iteration

DCGM_FI_DEV_GPU_UTIL is deliberately absent: it is the "utilisation" pinned at 99.9 % in every scenario, because a resident spin kernel counts as busy. Nothing in the loop is fitted to it.

THE QUESTION THAT MATTERS MOST

Where does "occupancy" come from on a real GPU?

The simulator's `sm_occupancy_pct` means the share of SM capacity retiring solver work over the sample window. No single counter reports that: three instruments, three scales, one composition rule.

fleet DCGM profiling metrics

Per-GPU counters at ~1 Hz, ~1 % overhead, always on (`dcgmi dmon`, DCGM exporter).

`SM_ACTIVE` : cycles with a warp resident.

`SM_OCCUPANCY` : resident warps / max.

`PIPE_FP64/FP32/TENSOR_ACTIVE` : cycles the pipe retired work. `DRAM_ACTIVE`,

`NVLINK_TX/RX`.

Role: the production analogue of the simulator's occupancy; what the detector runs on.

LAB Nsight Compute (ncu)

Per-kernel replay with full counters; tens of × overhead, so once per application on one node, never in production. `sm_warps_active` (achieved occupancy), `sm_throughput`, `dram_throughput`.

Role: fits the **work** → **occupancy** curve from 5–6 subdomain sizes, and says whether the solver is FP64-, memory- or latency-bound, which picks the DCGM pipe that stands in for "work".

job Nsight Systems (nsys) + NVTX

Low-overhead timeline of kernels and NCCL / MPI calls; NVTX ranges tag `HALO`, `COMPUTE`, `ALLREDUCE`, `OUTPUT` in the application.

Role: the real-fleet `rank_performance` : per-rank phase durations per iteration. The only way to see the barrier, and it shows the spin kernel that makes `GPU_UTIL` lie.

Composition rule for the fleet signal

Take the pipe the lab run says carries the work (FP64 for an implicit CFD solver). Per sample window: $\text{occupancy} \approx \text{mean}(\text{PIPE_FP64_ACTIVE})$, $\text{busy} \approx \text{mean}(\text{SM_ACTIVE})$, $\text{spinning} = \text{busy} - \text{occupancy}$. A rank parked at the barrier shows `SM_ACTIVE` near 1 and `PIPE_ACTIVE` near 0: the simulator's util-versus-occupancy gap. Its spin occupancy (0.05) is calibrated from that gap on healthy iterations.

Fitting the five arrows, one experiment each

Every experiment is a sweep a fleet can run on one node in an afternoon. Each fits the constants of exactly one arrow, so a bad fit points at one equation.

Arrow	Experiment	Fit	Expected shape
work → occupancy	Solver on one GPU at 6 subdomain sizes (30 k – 8 M cells); ncu per kernel, DCGM alongside	$c^{1/2}$ by least squares on $occ = n / (n + c^{1/2})$; also seconds per cell update from wall time / cells	saturation curve; DCGM pipe-active tracks ncu achieved occupancy within a few %
occupancy → power	Lock clocks (<code>nvidia-smi -lgc</code>) at 5 frequencies × 3 subdomain sizes, 60 s each; read <code>POWER_USAGE</code>	P_{idle} from the idle GPU; exponent by log–log slope of $(P - P_{idle})$ vs f at fixed occupancy; P_{max} at base clock, full occupancy	straight line in log–log with slope near 2–2.5; linear in occupancy at fixed f
power → temperature	Step test: idle → full load → idle, 5 min each; <code>GPU_TEMP</code> at 1 Hz, inlet from the rack sensor, rack kW from the PDU	τ from the exponential rise; $R_{th} = (T_{ss} - T_{inlet}) / P$; rack coupling from inlet vs rack kW over a day	one time constant fits the rise and the fall; residual white
temperature → throttle	No sweep: read <code>SLOWDOWN_TEMP</code> , <code>SHUTDOWN_TEMP</code> ; mine <code>CLOCK_THROTTLE_REASONS</code> transitions from history	engage threshold directly; hysteresis from engage–release temperature pairs; derate per °C from clock vs temperature while throttled	engage clusters at one temperature; release a few °C lower
throttle → clock	Read <code>SUPPORTED_CLOCKS</code> ; take every throttled second in history	step size and floor directly; check every observed <code>SM_CLOCK</code> sits on the lattice	a staircase, never a ramp
the lag	Same step test, 1 Hz	lagged cross-correlation of temperature crossing the threshold vs the first clock drop	peak at +1 sample: the governor acts on the next interval

Six things that must be true of real telemetry before any label exists

These test the shape of the loop, not its numbers. Each is a scripted check over ordinary healthy production days; a failure names the missing physics rather than a constant to tune.

#	Claim the loop makes	Check on fleet data	Pass looks like
1	Power follows work, not busy time	Regress <code>POWER_USAGE</code> on <code>PIPE_ACTIVE</code> and separately on <code>GPU_UTIL</code>	R^2 far higher for pipe-active; the <code>GPU_UTIL</code> fit is flat
2	Temperature is first-order in power	Fit the step-test τ to every natural load ramp in history (job starts)	one τ across GPUs, no second time constant
3	Inlet is a rack property	PCA of the 32 GPU temperatures in a rack after removing each GPU's own power term	first component carries most variance and tracks the inlet sensor
4	Throttling is a quantised staircase with hysteresis	Histogram of <code>SM_CLOCK</code> during throttled seconds; engage vs release temperatures	mass on the 15 MHz lattice; release 2–4 °C below engage
5	The governor acts one interval late	Lagged correlation of threshold crossings vs clock drops	peak at +1 sample, not 0
6	The barrier identity	From NVTX timing: per rank, halo + compute + wait + output = iteration time; wait of fast ranks = slowest rank's arrival – own arrival	identity holds to the ms; utilisation stays ~100 % while pipe-active drops on waiting ranks

Only after 1–6 pass is a scenario worth replaying. If check 1 fails, the workload is not occupancy-bound and the power arrow needs a bandwidth term; if 3 fails, cooling is per node or per aisle and the thermal domain is not the rack; if 6 fails, the job is not bulk-synchronous and the whole rank-spread family of signatures is off the table.

Replay the scenarios on the fleet, controlled first, then historical

Each simulated scenario has a safe real counterpart that can be staged on a drained node or rack. Score the simulator's telemetry and the diagnosis against what the fleet recorded.

Scenario	Controlled real counterpart	What to compare
thermal	CRAC set-point raised on one rack during a maintenance window, or a blanked cold-aisle tile	inlet ramp, 32-GPU co-movement, time to first throttle, staircase depth
gpu degradation	<code>nvidia-smi -lgc</code> cap at 90 % on one GPU of a running job	occupancy drop on the victim, rank spread, throttle-reason bit, peers' spin signature
network domain	Administratively down 7 of 8 uplinks on one leaf; keep one	queue depth, drops, halo time on cross-rack ranks only, compute unchanged
straggler	A CPU-side process pinned to one host stealing cycles from the launch thread	intermittent pacing duty, near-zero wait on the victim, no throttle bit
phase change	Turn checkpoint interval from 100 to 20 iterations in the job's own config	storage latency stays up, io pressure, rank spread flat, no device channel moves

Then history: every ticketed incident with a known cause, replayed through the same 100-iteration windowing and scored the way the simulator is scored, verdict and localisation against the ticket.

What counts as passing

Not numbers to the percent: the order of events, which channels move and which stay flat, and the diagnosis reaching the ticket's verdict and rack or GPU.

What a failure means

A signature the fleet shows and the simulator does not is a missing model term, never a constant to re-tune after stage 1 has fitted it.

Boundary to state up front

The simulator stops at throttling. A real cooling failure ends in thermal shutdown and a manual restart; that state is not modelled and no replay is scored past it.

IF THE ASSUMPTIONS CHANGE

What moves in the loop under a different workload or a different failure

The cycle is the invariant. A new workload changes the constants and the pipe that stands in for work; a new failure changes where the fault enters the cycle.

Different workload

Assumption	What changes
ML training instead of CFD	work pipe becomes <code>PIPE_TENSOR_ACTIVE</code> ; occupancy curve refitted per model; all-reduce is bandwidth-bound (gradients, GB not bytes), so the barrier arrow gains a size term
Memory-bound solver	power arrow gains a <code>DRAM_ACTIVE</code> term; the clock exponent falls because HBM does not scale with SM clock
Asynchronous or pipelined communication	the barrier identity weakens; rank spread stops being the headline diagnostic, straggler signatures blur
Subset allocations, mixed tenants	fleet-relative features must be computed over the job's GPUs only; idle neighbours are the most common thing in the data

Different failure

Failure	Where it enters the loop
Power cap (facility or board)	between power and clock: <code>THROTTLE_REASONS</code> power bit, clock falls with temperature flat; a new throttle reason, same staircase
HBM ECC storm	at occupancy: row remapping steals bandwidth; like gpu degradation but with <code>XID</code> / ECC counters as the witness
NIC flap, single optic	outside the loop, at the halo: per-port asymmetry the current fabric model cannot show; first roadmap item
Shared-aisle cooling	thermal domain becomes several racks; check 3 on the previous slide is what detects it
Thermal shutdown	past the loop's boundary: an absorbing state that needs an availability model, not a telemetry one

The test for any new assumption is the same three stages: fit its constants from a sweep, check its structure on healthy days, replay its fault under control. The arrows that did not change keep their fits.